DGMS 2019 • SECTION K: SOIL MECHANICS & ENGINEERING GEOLOGY

Question 82: Atterberg Limits & Soil Consistency: Plastic Limit Boundary

Official Examination Guide • Soil Consistency States, Phase Transitions & Geotechnical Indices

OFFICIAL EXAMINATION QUESTION

The Atterberg limits that define the boundary between the plastic and semi-solid states is

(a) Liquid limit

(b) Plastic limit CORRECT ANSWER

(c) Shrinkage limit

(d) Zero% water content and plastic limit

1. Executive Summary & Core Principle

In geotechnical engineering and soil mechanics (established by Swedish chemist **Albert Atterberg in 1911** and standardized by **Arthur Casagrande in 1932/1948**), fine-grained cohesive soils (clays and silts) exhibit four distinct physical states of consistency depending on their moisture content (w):

1. **Liquid State:** Soil behaves essentially as a viscous liquid, possessing negligible shear strength and flowing under its own weight or minimal shock.
2. **Plastic State:** Soil can be deformed and molded into various shapes without developing tensile cracks or crumbling.

3. **Semi-Solid State:** Soil can support substantial loads, but when deformed, it fractures, shears, and crumbles rather than deforming plastically. Further reduction in moisture produces a corresponding reduction in total soil volume.
4. **Solid State:** Soil volume remains completely constant; further drying causes air to replace water in the pores with zero shrinkage.

The boundaries between these four states are quantified by critical water contents known as the **Atterberg Limits**:

- **Liquid Limit (w_L or LL):** The boundary between the *Liquid state* and the *Plastic state*.
- **Plastic Limit (w_P or PL):** The boundary between the **Plastic state and the Semi-Solid state**.
- **Shrinkage Limit (w_S or SL):** The boundary between the *Semi-Solid state* and the *Solid state*.

Therefore, the Atterberg limit defining the boundary between the plastic and semi-solid states is unequivocally the **(b) Plastic limit**.

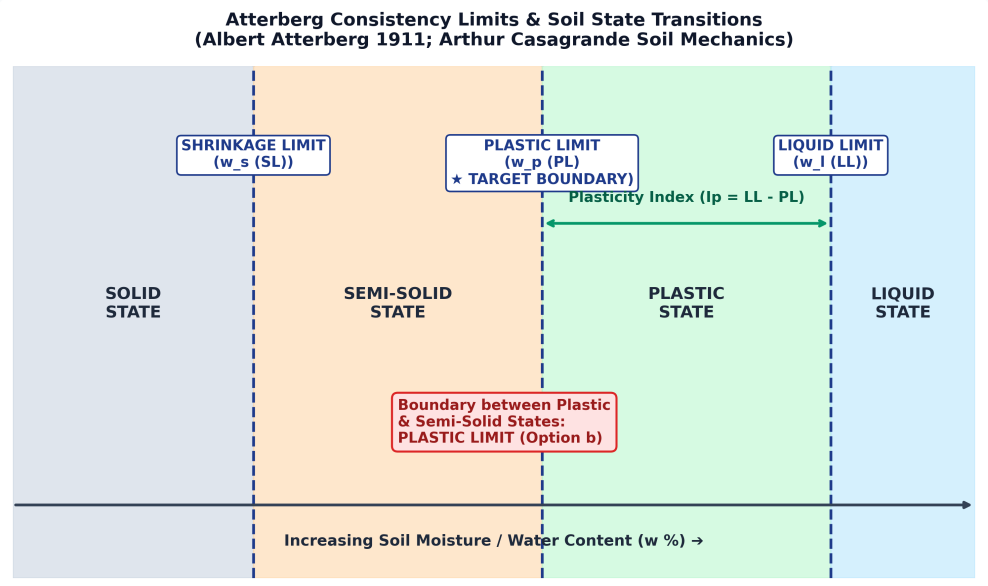


Figure 82.1: Phase diagram of fine-grained soil consistency states and boundary water contents (Atterberg Limits), demonstrating that the Plastic Limit (w_P) defines the exact transition between the plastic and semi-solid states (Grounded in Terzaghi & Peck and Casagrande).

2. Laboratory Determination of the Atterberg Limits

Atterberg Limit	Boundary Demarcated	Standard ASTM / IS Laboratory Method	Engineering Significance
Liquid Limit (w_L / LL)	Liquid State $ \leftrightarrow$ Plastic State	Casagrande Cup (moisture at which standard groove closes along 12.7 mm after 25 drops) or Fall Cone Penetrometer (20 mm cone penetration).	Evaluates soil compressibility ($C_c \approx 0.009(\text{LL} - 10)$) and settlement potential.
Plastic Limit (w_P / PL)	**Plastic State $ \leftrightarrow$ Semi-Solid State**	Rolling soil into a thread on a glass plate; moisture content at which the thread crumbles precisely at **3.2 mm (1/8 inch) diameter**.	Determines the lower threshold of plasticity and workability of cohesive soil.
Shrinkage Limit (w_S / SL)	Semi-Solid State $\leftrightarrow$ Solid State	Mercury displacement method; moisture content below which drying produces no further volume decrease ($\Delta V = 0$).	Evaluates swelling/ shrinkage potential in expansive clay soils (e.g., Black Cotton Soils / Smectites).

3. Derived Geotechnical Consistency Indices

From the fundamental Atterberg limits, several crucial engineering indices are calculated:

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Plasticity Index (Ip or PI):
Ip = wL - wP (Range of moisture over which soil exhibits plastic behavior)

Liquidity Index (IL):
IL = (w - wP) / (wL - wP) = (w - wP) / Ip
• IL < 0 : Semi-solid or solid state (brittle failure risk)
• 0 < IL < 1 : Plastic state (normal field consolidation)
• IL > 1 : Liquid state (sensitive clay quick failure hazard)

Consistency Index (IC):
IC = (wL - w) / Ip
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4. Systematic Option Evaluation

- **(a) Liquid limit: INCORRECT.** Defines the boundary between the *liquid state* and the *plastic state*.
- **(b) Plastic limit: CORRECT ANSWER.** By definition, the plastic limit marks the moisture boundary separating the **plastic state** from the **semi-solid state**.
- **(c) Shrinkage limit: INCORRECT.** Defines the boundary between the *semi-solid state* and the *solid state*.
- **(d) Zero% water content and plastic limit: INCORRECT.** Zero% moisture represents an oven-dry state deep within the solid state, not a boundary of the plastic state.

> **Key Takeaway:** Liquid Limit = Liquid/Plastic boundary; Plastic Limit = Plastic/Semi-solid boundary; Shrinkage Limit = Semi-solid/Solid boundary. Hence, **(b) Plastic limit** is the correct answer.

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